# PRASANNA REDDY PULAKURTHI

Email: pp4405@rit.edu | Phone: 585-355-6034 | Location: San Jose, CA | LinkedIn: prasannapulakurthi | Website: prasannapulakurthi.com

## Education

**Rochester Institute of Technology** — Rochester, NY
*Doctor of Philosophy (Ph.D.) – Electrical and Computer Engineering* | ***GPA: 3.93/4.0*** — *Aug 2019 to Aug 2025 (expected)*

**Rochester Institute of Technology** — Rochester, NY
*Master of Science (MS) – Electrical Engineering* | ***GPA: 4.0/4.0*** — *Aug 2017 to Jul 2019*

**PESIT Bangalore South Campus** — Bangalore, India
*Bachelor of Engineering (BE) – Electronics and Communications Engineering* — *Aug 2013 to Jun 2017*

## Technical Skills

- **LANGUAGES & SOFTWARE TOOLS**: C, Python, HTML/CSS, MATLAB, Linux, ROS, Gazebo, Docker, LaTeX, Git, SQL, Blender
- **AI/ML**: PyTorch, TensorFlow, TensorBoard, Keras, CUDA, GPU, PyTorch DDP, TensorRT, ONNX, Scikit-Learn, OpenCV, spaCy, mmCV, Hugging Face Transformers, LLM, NLP, MMAction2, SAM2, DINOv2, Path Planning, ORB-SLAM, CLIP, OpenAI, BLIP-2, LLaVA, Qwen-VL, MLLM

## Professional Experience

**Rochester Institute of Technology**
*Ph.D. Researcher, ML & Computer Vision* — *May 2024 to Aug 2024*

- Developed two novel **Data Augmentation** techniques (1. Dual-region and 2. Shuffle PatchMix) to improve model robustness and generalization. Demonstrated improvements in performance on domain adaptation datasets such as PACS by up to **+7.3 pp** (percentage points), and person re-identification datasets such as Market-1501 by up to **+6.91 pp**.

**DEVCOM Army Research Laboratory (via Booz Allen Hamilton and MBO Partners)**
*Perception Engineer Intern* — *May 2023 to Aug 2023*

- Advanced Generative Models: Developed a GAN variant combining neural-architecture search, PMish activation, bounded MMD repulsive loss, and adaptive rank decomposition, achieving a reduction of FID scores by **34%** (CIFAR-10) and **28%** (STL-10) while compressing the generator **4–10**×, enabling real-time edge deployment.
- Data Generation: Generated new synthetic gesture video data for Human Action Recognition using Generative Adversarial Network and significantly lifted human-action recognition accuracy across ground (**+4.4 pp**) & aerial (**+7.7 pp**) views.

**DEVCOM Army Research Laboratory (via Booz Allen Hamilton and MBO Partners**
*Perception Engineer Intern* — *May 2022 to Aug 2022*

- Developed domain adaptation techniques to improve model robustness across different digit domains, lifting cross-domain accuracy by up to **+2.56 pp** on difficult shifts.
- Globally-optimal Iterative ML Classifier: Formulated a fast, globally-optimal Iterative Maximum Likelihood Classifier by deriving a fixed-point solver from a regularized ML objective, achieving ∼3× faster convergence compared to gradient descent.

**Qualcomm Internship – Virtual Reality (VR) Team**
*Engineering Intern* — *May 2021 to Aug 2021*

- Developed a novel temperature-compensation algorithm enhancing the accuracy of the 6DOF 4Tx ultrasound VR controller.
- Implemented the algorithm in C, improving the root mean squared error (RMSE) by **+11.29 pp** on synthetic log data.

**Microsoft Internship – Iris Recognition Team for HoloLens 2 Augmented Reality (AR) Team**
*Engineering Intern* — *May 2020 to Aug 2020*

- Optimized Gabor filter frequencies in Iris Recognition pipeline, significantly enhancing signature uniqueness.
- Developed GAN-based Super Resolution to enhance low-resolution Iris image quality while retaining key Gabor frequencies.

**Oak Ridge National Laboratory [Link]**
*Research Assistant* — *May 2018 to Jul 2019*

- Designed and executed an unmanned aerial system (UAS) data collection to capture aerial imagery of a building casting shadows on 14 differently colored panels, both in and out of shadow.
- Developed six algorithms for shadow detection in aerial images using color transforms and machine learning techniques, leveraging collected data. Improved upon traditional approaches by **+1.51%** in accuracy and **+2.4%** in F-score.

## Select Publications

- **Prasanna Reddy Pulakurthi** et al. "X-CoT: Explainable Text-to-Video Retrieval via Large Language Models (LLM) based Chain-of-Thought Reasoning," *EMNLP* 2025.
- **Prasanna Reddy Pulakurthi** et al. "Shuffle PatchMix Augmentation with Confidence Margin Weighted Pseudo Labels for Enhanced Source-Free Domain Adaptation," *IEEE ICIP*, 2025. [Link]
- **Prasanna Reddy Pulakurthi** et al. "Effective dual-region augmentation for reduced reliance on large amounts of labeled data," *Synthetic Data for Artificial Intelligence and Machine Learning: Tools, Techniques, and Applications III*. SPIE, 2025. [Link]
- **Prasanna Reddy Pulakurthi** et al. "Enhancing GANs with MMD Neural Architecture Search, PMish Activation Function and Adaptive Rank Decomposition," *IEEE Access*, 2024. [Link]
- **Prasanna Reddy Pulakurthi** et al. "Enhancing human action recognition with GAN-based data augmentation," *Synthetic Data for Artificial Intelligence and Machine Learning: Tools, Techniques, and Applications II*. SPIE, 2024. [Link]
- **Prasanna Reddy Pulakurthi** et al. "Enhancing GAN Performance through Neural Architecture Search and Tensor Decomposition," *IEEE ICASSP*, 2024. [Link]

## Projects

**Machine Learning and AI Projects**

- Applied Generative Adversarial Networks (GANs) to unsupervised domain adaptation, image-to-image translation, and human action recognition tasks. [Link]
- Developed Invertible Neural Networks (INNs) to perform super-resolution and image denoising. [Link]

**Vision Captioning Assistant (V-CAT) [Link]**

- Built a full-stack, camera-driven “smart-glasses” assistant that streams live video to vision–language models and returns real-time scene captions plus synthesized speech for hands-free accessibility.

**Signmentor AI (ASL Sign Tutoring) [Link]**

- Built a full-stack website for ASL sign tutor that accurately identifies every handshape in the ASL alphabet (A to Z) by pairing your camera feed with cutting-edge Multi-modal LLM for fast, reliable classification and instant learner feedback.

**Comparison of Simultaneous Localization and Mapping (SLAM) Algorithms in ROS Environment [Link]**

- Simulated and benchmarked Gmapping (2D laser-based), Octomap (probabilistic 3-D occupancy-grid), and RTAB-Map (RGB-D dense mapping) in a Gazebo/ROS indoor environment using custom and stock robots outfitted with LiDAR and RGB-D sensors.
- Evaluated map quality, runtime, and sensor cost, recommending Gmapping for lightweight real-time 2-D mapping, Octomap for memory-efficient 3-D obstacle maps, and RTAB-Map for high-fidelity textured reconstructions.

**Image and Signal Processing Projects**

- Implemented several image and signal processing algorithms, such as image resizing, noise filtering, image enhancement, image segmentation, camera calibration, stereo reconstruction, affine, and projective transforms.

**Real-Time Attendance System using Image Processing [Link]**

- Developed a real-time automatic attendance system using OpenCV for face detection, Python-implemented Binarized Statistical Image Features for face recognition, and a Kivy-based app interfacing via sockets and PHP with an Apache-hosted GUI displaying attendance data stored in MySQL.

**Automatic License Plate Detection and Recognition [Link]**

- Built an automatic number plate detection and recognition system using edge and histogram-based image processing methods for detection and neural networks for recognition, coded in MATLAB.

## Coursework

- **Signal & Image Processing**: Digital Signal Processing (EEE678), Digital Image Processing (EEE779), Multi-view Imaging (IMGS712)
- **Machine Learning**: Deep Learning Independent Study (EEE799), AI Explorations (EEE 647), Pattern Recognition (EEE670)
- **Communications**: Information Theory Coding (EEE794), Image and Video Compression (EEE781), Adv. Engineering Mathematics
- **Machine Learning Course**: Machine Learning by Stanford University on Coursera